

Course 3 Mentor Synchronization - RC1 QA Report

Reconciled 140-lesson beginner-first Mentor | 11 September 2026

Result	RC1 QA PASS WITH 1 EXTERNAL PWA/DEVICE BOUNDARY
Automated controls	60 PASS 1 PENDING 0 FAIL
Curriculum	10 phases 140 exact lessons C3-110 through C3-116 final layer
Storage	daMentorDataEngineering.v1 preserved; legacy migration tested

What was synchronized

The Mentor now routes every lesson to the reconciled RC1 teaching book page, matching practice page, protected review page and starter asset where present. Phase Start Here, setup, Mini-Lab, Gate case and remediation files are wired from the rebuilt phase packages. C3-112 reviewer material is deliberately absent from the learner app.

Defects caught during real browser QA

- 1. Learn-page HTML nesting defect.** A missing closing DIV made steps 3-8 become grid children of step 2 on mobile, causing overlap and click interception. The DOM is repaired; all eight teacher steps are now direct siblings.
- 2. Disabled progression-state defect.** Guided state saved but the Learn view did not re-render, so Save genuine attempt stayed disabled until navigation. The state path now re-renders immediately; Guided -> Attempt -> Review -> Fresh retry works in place.

Core regression evidence

Static Mentor sync	16/16 PASS
Routed assets	176/176 valid local headers
Exact PDF page targets	420/420 in range
Routed ZIP CRC	46/46 PASS
HTTP direct-open	181/181 HTTP 200
Chromium DOM/state/layout	20/20 PASS
Legacy migration + backup/restore	7/7 PASS
JS / PWA static controls	11 PASS + 1 external PENDING

State and assessment behavior

Attempt protection

Review is locked before a genuine attempt, then unlocks after the attempt; changed retry remains locked until review.

Gate retests

A failed Gate A does not reveal Gate B. An explicit full-retest assignment is required. Gate C remains sealed until separately assigned.

Final graduation

C3-113 uses 18 boolean controls: 8 competencies plus required execution/proof controls. There are no numeric final competency inputs, percentage passes or compensating averages.

Legacy safety

Old numeric competency values cannot satisfy the strict final helper. A legacy finalGatePassed flag is cleared unless the new strict evidence contract is actually met.

Final-layer isolation

C3-104 remains the C3-09 phase Gate. C3-113 is the separate course-wide graduation Gate. C3-112 remains reviewer-only.

Progress continuity

Existing storage key is preserved. Legacy lesson, Gate, Bridge-entry, notes, study sessions and evidence migration was simulated successfully. Current-schema backup/restore round-trip also passed.

Responsive and theme QA

Real headless Chromium showed no document-level horizontal overflow at 360, 390, 412, 768 or 1440 px. AMOLED resolves to true #000. The repaired 390 px Learn screen was visually inspected after DOM repair; teacher steps, buttons and status card are clean.

PWA/runtime boundary

Service-worker JavaScript, cache version, network-first/no-store policy, manifest, core cached assets and app scripts pass static QA. This managed Chromium environment blocks navigation to localhost/private origins and file URLs by administrator policy, so real service-worker registration/install cannot be independently exercised here. Physical Android install/persistence/background behavior also remains device evidence. These are deployment/device boundaries, not hidden PASS claims.

Route and package integrity

The synchronized Mentor contains the current reconciled learner-facing assets rather than the old INTERNAL_QA bundle. Routes were validated against actual files and actual PDF page counts.

Measure	Verified
Course phases	10
Canonical lesson routes	140
Lesson/practice/review page targets	420
Unique routed learner assets	176
HTTP app + route assets	181
Routed ZIP archives	46
Reviewer C3-112 learner leakage	0
Required external media dependency	0

Release interpretation

Mentor package QA: PASS. The learner app, state logic, assessment topology, route files, mobile layout, migration and package assets are synchronized and regression-tested.

Hosted/PWA/device evidence: PENDING external environment verification. The package does not claim what this managed browser cannot execute.

Heavy engineering runtimes: Course artifacts intentionally require real learner/runtime evidence for dbt, Airflow, Spark, Kafka, Delta, Fabric and protected GitHub/OIDC behaviors where those skills are assessed. Artifact QA never substitutes for those execution proofs.